Bayesian neural-network learning with non-Gaussian weight priors

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

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Abstract

Can a Bayesian neural network learn a small circuit when its independent weight priors are not Gaussian? We prove that three width-independent learning guarantees survive for a natural family of non-Gaussian coordinate laws: even densities $Z^{-1}e^{-V}$, with variance one and curvature bounded above and below by positive constants. The guarantees cover the strong-prior limit, taken separately at each fixed width, and the finite-scale regimes $\gamma > 1$ and $\gamma = 1$, in every case at every width at least the target's bounded-weight network complexity. The sample bound is polynomial in input dimension and target complexity at each fixed depth, independently of the learner's width. The model uses tanh hidden layers, a linear readout, an absolute output margin imposed at every Boolean input before training, and exact conditioning on noiseless labels. The prediction guarantee controls the actual posterior, including mixtures: its correct-label probability exceeds 2/3 on almost all test inputs. The proof uses a translation-entropy bound for target mass and concentration of strongly log-concave measures. At mean-field scaling, a relative Lindeberg estimate controls log partition functions for finite covariance tests with polynomial losses; an ordinary central limit approximation would not provide the needed bound. We also give a broader quadratic-tail extension of the strong-prior result. No efficient posterior computation or polynomial dependence on depth is asserted.

1 Introduction

A useful sample bound for a large Bayesian neural network should depend on the size of a small implementation of the target, rather than on the number of parameters available to the learner. Gaussian weight priors permit such bounds in several small-scale and mean-field regimes. This paper establishes that exact Gaussianity is unnecessary for three of those guarantees.

Our main family consists of independent coordinates with a fixed even density $q(t) = Z^{-1}e^{-V(t)}$, standardized to variance one, where

$$V \in C^2(\mathbb{R}), \qquad 0 < m \le V''(t) \le M < \infty \quad (t \in \mathbb{R}). \tag{1}$$

The lower bound supplies concentration. The upper bound makes a translated copy of a small target inexpensive in relative entropy. These are restrictions on a one-dimensional weight law, rather than restrictions selected separately for each network function. A genuinely non-Gaussian example is the variance-standardized law with density proportional to $\exp\{-t^2/2 - b \log\cosh t\}$, for any fixed $b > 0$.

*Authors are listed in alphabetical order. The main learning statements have Lean proofs; see the repository for scope and exclusions.

We retain the same layer scales and the same learning rule throughout. Before training, the prior is conditioned to produce outputs of absolute value greater than $1/2$ at every Boolean input. Training then conditions on exact agreement with noiseless labels. At each fixed depth, we prove a sample bound polynomial in dimension and bounded-weight target complexity, uniformly over every learner width at least that complexity. There are three regimes: a common scale tending to zero at fixed width; finite-scale priors with exponent $\gamma > 1$; and the mean-field endpoint $\gamma = 1$. The latter two statements require neither a small-scale limit nor an infinite-width limit.

The three proofs use a shared statistical principle. A finite collection of simply describable functions contains all but a sufficiently small amount of prior mass relative to the target. A union bound eliminates its inaccurate members after polynomially many examples. In the strong-prior limit, quadratic tail envelopes select functions of bounded weighted squared norm. For $\gamma > 1$, concentration permits a spectral truncation with the same effect. At $\gamma = 1$, the bulk internal matrices cannot be removed that way. We instead approximate empirical hidden-layer covariances by controlled exponential tilts, and then realize those covariances by small finite networks.

The new difficulty at the mean-field endpoint is that non-Gaussian internal rows are not determined by their covariance. We resolve it with a relative Lindeberg comparison for log partition functions. Its error is polynomial in the tilt budget and independent of the number of input points. This avoids both an exponential dependence on the tilt budget and a factor equal to the size of the Boolean cube. All supporting compression and posterior arguments are included below and in the appendices.

2 Model and guarantees

Let $d \geq 1$, $L \geq 2$, and $\mathcal{X} = \{-1, 1\}^d$. Write $\bar{x} = (1, x_1, \ldots, x_d)$. A depth-$L$ network has $L - 1$ hidden layers of common width N and computes

$$h^0(x) = \bar{x}, \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \ (1 \leq s < L), \qquad z_w(x) = W^L h^{L-1}(x).$$

There are no separate internal biases. For a Boolean function g, let

$$U_{N,g} = \{w : g(x) z_w(x) > 1/2 \text{ for every } x \in \mathcal{X}\}, \qquad U_N = \bigcup_g U_{N,g}.$$

If μ is the independent-coordinate parameter prior, its induced function prior is $\pi(g) = \mu(U_{N,g})/\mu(U_N)$. All priors used below have positive densities, so any nonempty strict-margin region has positive mass. In particular the learner is defined whenever the target has a finite implementation at the chosen width.

Let $C_L(f)$ be the least integer $C \geq 1$ for which f has a width-C, depth-L implementation with the stated strict margin and all weights at most C in absolute value. This combines width and weight magnitude at the specified depth; it is not an unrestricted gate count. Given independent inputs $x_1, \ldots, x_n \sim D$ and labels $f(x_i)$, the posterior is

$$\Pi(g \mid S_n) = \frac{\pi(g)\mathbf{1}\{g(x_i) = f(x_i) \ \forall i\}}{\sum_h \pi(h)\mathbf{1}\{h(x_i) = f(x_i) \ \forall i\}}.$$

Define its Gibbs risk and confidence-failure probability by

$$R_D(\Pi) = \mathbb{E}_{g\sim\Pi}\mathbb{P}_{x\sim D}\{g(x) \neq f(x)\}, \qquad F_q(\Pi) = \mathbb{P}_{x\sim D}\{\Pi(g(x) = f(x) \mid S_n) \leq q\}.$$

For every posterior,

$$F_q(\Pi) \leq \frac{R_D(\Pi)}{1 - q}. \tag{2}$$

Thus the statements below control the entire posterior, rather than a selected minimizer or the mean of its real-valued outputs. Posterior-majority error is at most $2R_D(\Pi)$, including arbitrary tie breaking.

Let ξ have density satisfying (1), be even, and have variance one. Independent copies are used for all scalar weights. The *strong-prior scales* are

$$W_{ij}^1 = \sigma\sqrt{N/d}\,\xi_{ij}^1, \qquad W_{ij}^s = \sigma\xi_{ij}^s \ (2 \le s < L), \qquad W_j^L = \sigma N^{-1/2}\xi_j^L. \tag{3}$$

The *finite-scale exponent family* is

$$W_{ij}^1 = N^{(1-\gamma)/2}\xi_{ij}^1/\sqrt{d}, \qquad W_{ij}^s = N^{-\gamma/2}\xi_{ij}^s \ (2 \le s < L), \qquad W_j^L = N^{-(\gamma+1)/2}\xi_j^L. \tag{4}$$

These are exactly (3) at $\sigma = N^{-\gamma/2}$. At $\gamma = 1$, the first-layer, internal, and readout variances are $1/d$, $1/N$, and $1/N^2$.

Theorem 2.1 (Strong-prior learning)**.** *Fix L and a coordinate law as above. There is a polynomial p_L, with coefficients allowed to depend on the law, such that the following holds. Let $C = C_L(f) < \infty$, $N \ge C$, and $\varepsilon \in (0,1)$. There is $\sigma_0 > 0$ such that for every $0 < \sigma < \sigma_0$, every input distribution D, every $\delta \in (0,1)$, and*

$$n \ge \frac{6}{\varepsilon}\left[p_L(d, C) + \log\frac{1}{\delta}\right],$$

with probability at least $1-\delta$, $R_D(\Pi) \le \varepsilon/3$ and $F_{2/3}(\Pi) \le \varepsilon$. The sample polynomial is independent of N. The threshold σ_0 may depend on the finite width, target, and desired accuracy; no uniform small-scale threshold over all widths is asserted.

Theorem 2.2 (Finite-scale learning for every $\gamma \ge 1$)**.** *Fix L, a coordinate law as above, and $\gamma \ge 1$. There is a polynomial $p_{L,\gamma}$ such that, for every $C = C_L(f) < \infty$, every $N \ge C$, every distribution D on $\mathcal{X}$, and every $\varepsilon, \delta \in (0,1)$, the condition*

$$n \ge \frac{6}{\varepsilon}\left[p_{L,\gamma}\left(d, C, \log\frac{1}{\varepsilon}\right) + \log\frac{6}{\varepsilon\delta}\right] \tag{5}$$

implies, with probability at least $1-\delta$,

$$R_D(\Pi) \le \varepsilon/3, \qquad F_{2/3}(\Pi) \le \varepsilon.$$

This is a finite-width statement for the actual non-Gaussian prior (4). The polynomial is independent of N, but its degree and constants may depend on L, γ, and the coordinate law. No uniformity as $\gamma \downarrow 1$ is claimed for the $\gamma > 1$ proof; the endpoint has its own proof.

Theorems 2.1 and 2.2 are noiseless, statistical statements. They do not provide efficient posterior algorithms, useful optimized numerical exponents, or polynomial dependence on depth. A different fixed confidence threshold in $(0,1)$ follows from (2) by changing constants.

3 Two elementary posterior estimates

Lemma 3.1 (A finite relevant class)**.** *Suppose a prior on finitely many functions assigns positive mass to f. Let $\mathcal{H}$ contain f, with $\log|\mathcal{H}| \le H$ and $\pi(\mathcal{H}^c)/\pi(f) \le \tau$. If $n \ge a^{-1}[H + \log(1/\delta)]$, then its consistency posterior has risk at most $a + \tau$ with sample probability at least $1-\delta$.*

Proof. Each function of true error greater than a survives all n labels with probability at most e^{-na}. A union bound excludes all such survivors in $\mathcal{H}$ with the stated probability. For every sample the posterior denominator is at least $\pi(f)$, so its mass outside $\mathcal{H}$ is at most τ. □

Lemma 3.2 (Learning from target mass)**.** *If $-\log \pi(f) \le B$, then $n \ge a^{-1}[B + \log(1/(\tau\delta))]$ implies posterior risk at most $a + \tau$ with sample probability at least $1 - \delta$.*

Proof. Let Z be posterior mass on functions of error greater than a. The target lower-bounds the posterior denominator, whence

$$\mathbb{E}Z \le \pi(f)^{-1} \sum_{g:\mathrm{err}_D(g)>a} \pi(g)(1 - \mathrm{err}_D(g))^n \le e^{B-na}.$$

Markov's inequality gives $Z \le \tau$ with probability at least $1 - \delta$. All other functions have error at most a. □

4 A target-mass bound at every width

Throughout this section ν has the even density in (1). Adding a constant to V changes nothing, so take $V(0) = 0$. Evenness gives $V'(0) = 0$ and hence $mt^2/2 \le V(t) \le Mt^2/2$.

Lemma 4.1 (Concentration and translation)**.** *A product of ν coordinates, conditioned on any centrally symmetric convex set of positive probability, is centered and satisfies*

$$\mathbb{E}e^{\lambda u \cdot Z} \le \exp\{\lambda^2 \|u\|_2^2/(2m)\}. \tag{6}$$

For the unconditioned product measure μ, Euclidean enlargement obeys

$$\mu(A_r) \ge \Phi(\Phi^{-1}(\mu(A)) + \sqrt{m}\, r). \tag{7}$$

Finally, if E is any centrally symmetric event of positive probability and a is deterministic, then

$$\mu(E + a) \ge \mu(E) \exp\{-M\|a\|_2^2/2\}. \tag{8}$$

Proof. Conditioning on a convex set preserves m-strong log-concavity. The Brascamp–Lieb variance inequality [2], applied after any linear exponential tilt, bounds the Hessian of its log moment generating function by $m^{-1}I$. Central symmetry makes the first derivative at zero vanish. Integrating twice proves (6). Convex indicator potentials follow by approximation with smooth convex barriers.

For (7), the Caffarelli contraction theorem [3] gives an $m^{-1/2}$-Lipschitz transport from standard Gaussian measure to μ. The inverse image of A_r contains the $\sqrt{m}\, r$ enlargement of the inverse image of A. Gaussian isoperimetry [4] gives the displayed inequality.

For the last assertion let Q be the law of $Z + a$ conditional on $Z \in E$. Direct calculation of relative entropy gives

$$D(Q\|\mu) = -\log \mu(E) + \mathbb{E}\left[\sum_i (V(Z_i + a_i) - V(Z_i)) \mid E\right].$$

Taylor's theorem bounds the sum by $\sum_i a_i V'(Z_i) + M\|a\|_2^2/2$. The conditional expectation of its linear term is zero because V' is odd and the conditional law is invariant under simultaneous negation. Since Q is supported on $E + a$, $D(Q\|\mu) \ge -\log \mu(E + a)$. This proves (8). □

Lemma 4.2 (One-dimensional small balls)**.** *Every centered even log-concave variable Y of variance at most $1/m$ satisfies*

$$\mathbb{P}\{|Y| \le \rho\} \ge c_m \rho \quad (0 < \rho \le 1), \qquad c_m = \tfrac{3}{8}\min\{1, \sqrt{m}\}.$$

This applies to every normalized sum of disjoint coordinates of Z.

Proof. Let $R = 2\max\{1, m^{-1/2}\}$. Chebyshev gives probability at least $3/4$ in $[-R, R]$. Symmetric unimodality gives probability at least $(\rho/R)(3/4)$ in $[-\rho, \rho]$. Log-concavity is preserved by convolution and linear projection, and normalized independent sums retain variance at most $1/m$. □

We first construct an exact clone of a target. Partition the N neurons of each hidden layer into C groups of sizes $k_a \in \{\lfloor N/C \rfloor, \lceil N/C \rceil\}$. Copy the corresponding teacher neuron throughout each group. For an internal teacher weight W_{ab} use W_{ab}/k_b on every edge from group b to group a. Divide the readout weight from group a by k_a. The cloned network computes exactly the same function. Let P project onto the group-constant subspace, with orthonormal basis $u_a = k_a^{-1/2}\mathbf{1}_{\text{group } a}$.

Define the weighted energy

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1}\|W^s\|_F^2 + N\|W^L\|_2^2, \qquad \kappa_N(g) = \inf_{w \in U_{N,g}} \kappa_N(w). \tag{9}$$

The first-layer clone contribution is at most $d(d+1)C^2$. Because $\max k_a / \min k_a \le 2$ and $N/\min k_a \le 2C$, its internal contribution is at most $2(L-2)C^4$ and its readout contribution at most $2C^4$. Amplify the teacher readout by three; the resulting exact clone has signed margin greater than $3/2$. A convenient common bound for its energy is

$$B_0 = d(d+1)C^2 + 2(L-2)C^4 + 36C^4. \tag{10}$$

The larger final constant leaves harmless slack.

Proposition 4.3 (Target mass)**.** *For each fixed L there is a polynomially bounded function $A_L(d, C)$, depending also on m, M, such that*

$$\mu_{N,\gamma}(U_{N,f}) \ge e^{-A_L(d,C)N^\gamma} \qquad (\gamma \ge 1,\ N \ge C = C_L(f)). \tag{11}$$

Proof. Put $t = N^{(1-\gamma)/2} \le 1$. In standardized coordinates impose the centered cylinder $\mathcal{C}_\rho$ consisting of the following constraints: all first-layer and readout coordinates have absolute value at most ρ; each internal row obeys $|\langle Z_{j,*}, u_a\rangle| \le \rho$ for every group a. These constraints involve independent random variables: within an internal row the sums concern disjoint groups. There are

$$r_0 = N[d + 2 + (L-2)C]$$

constraints. Lemma 4.2 gives $\mu(\mathcal{C}_\rho) \ge (c_m\rho)^{r_0}$.

Conditional on the cylinder, every internal matrix is centered and m-strongly log-concave. For unit vectors u, v, the scalar $u^T Z v$ is therefore $m^{-1/2}$-sub-Gaussian. Two $1/4$-nets, each with at most 9^N elements, give

$$\mathbb{P}\{\|Z\|_{\text{op}} > 2q\sqrt{N} \mid \mathcal{C}_\rho\} \le 2\exp\{(2\log 9 - mq^2/2)N\}.$$

Take

$$b_{m,L} = 2\sqrt{\frac{2}{m}(2\log 9 + \log(4L))}.$$

Each conditional failure probability for $\|Z\|_{\mathrm{op}} \le b_{m,L}\sqrt{N}$ is at most $1/(2L)$. Let E be the cylinder intersected with these bounds for all internal layers. It is centrally symmetric and has $\mu(E) \ge \frac{1}{2}(c_m\rho)^{r_0}$.

Translate this event by the amplified clone in standardized coordinates. The internal weight perturbation decomposes as

$$\frac{t}{\sqrt{N}}Z = R + E_\perp, \qquad R = \frac{t}{\sqrt{N}}ZP, \qquad E_\perp = \frac{t}{\sqrt{N}}Z(I-P),$$

where $\|R\|_{\mathrm{op}} \le \rho t\sqrt{C}$, $E_\perp P = 0$, and $\|E_\perp\|_{\mathrm{op}} \le b_{m,L}t$. We detail stability to make the role of the cylinder explicit. Write $e_s = N^{-1/2}\|h^s - h^s_*\|_2$, with h^s_* the exact clone's activations. Its internal matrices have norm at most $\sqrt{2}C^2$, and $h^s_* \in \mathrm{range}(P)$. The first layer has

$$e_1 \le \rho t(d+1)/\sqrt{d} \le \rho(d+1).$$

For $s \ge 2$, boundedness and the contraction property of tanh give

$$e_s \le (\sqrt{2}C^2 + b_{m,L})e_{s-1} + \rho\sqrt{C}.$$

Set $A_0 = 2 + 3C^2 + b_{m,L}$. Induction bounds $e_{L-1} \le \rho(d+L+1)A_0^{L-2}$. The cloned readout has $\sqrt{N}\|v_*\|_2 \le 3\sqrt{2}C^2$, and the perturbation readout has ℓ^1 norm at most $\rho t \le \rho$. Consequently the output error is at most

$$3\sqrt{2}C^2\rho(d+L+1)A_0^{L-2} + \rho.$$

Choose

$$\rho = [2 + 9C^2(d+L+1)A_0^{L-2}]^{-1}.$$

The last error is strictly smaller than $1/2$, so the translated event lies in $U_{N,f}$, indeed with signed margin greater than one.

If a is the standardized translation, then $\|a\|_2^2 = N^\gamma\kappa_N(w_*) \le B_0N^\gamma$. Lemma 4.1 gives

$$-\log\mu_{N,\gamma}(U_{N,f}) \le \tfrac{M}{2}B_0N^\gamma + N[d+2+(L-2)C]\log\frac{1}{c_m\rho} + \log 2.$$

Since $N \le N^\gamma$, one may take

$$A_L(d,C) = 1 + \tfrac{M}{2}B_0 + [d+2+(L-2)C]\log\frac{1}{c_m\rho} + \log 2. \tag{12}$$

At fixed depth and fixed curvature constants this has a polynomial majorant in d, C. □

5 The strong-prior limit and quadratic-tail envelopes

We first record the deterministic counting consequence of Appendix A.

Lemma 5.1 (A width-independent energy count)**.** *For every fixed L there is a polynomially bounded $H_L(d,K)$ such that*

$$\log\#\{g : \kappa_N(g) \le K\} \le H_L(d,K) \qquad (K \ge 1),$$

uniformly over all widths. The same bound holds for the union of these function classes over widths.

Proof. Take a representative of energy less than $K+1$ for each function. The compression theorem in Appendix A provides an implementation of width $b = m_L(d, K+1)$, strict margin $1/2$, and weights at most b, where b is polynomially bounded at fixed depth. Round each weight to a multiple of $(8\Lambda)^{-1}$, where

$$P = b(d+2) + (L-2)b^2, \qquad \Lambda = (d+Lb)[1+b(b+1)]^{L-1}.$$

A layerwise maximum-norm induction bounds the output change by $1/8$; this is also proved in Lemma C.3. Thus all signs survive. There are at most $(3+16b\Lambda)^P$ rounded arrays. Take $H_L(d,K) = P\log(3+16b\Lambda)$. This counts one common compressed family for every original width. □

The next statement permits many priors outside the curvature family.

Proposition 5.2 (Two-sided quadratic envelopes)**.** *In the strong-prior scales (3), replace standardized coordinates by independent densities q_j satisfying*

$$c_j e^{-bu^2} \le q_j(u) \le C_j e^{-au^2} \quad (u \in \mathbb{R}), \qquad 0 < a \le b < \infty, \tag{13}$$

where $c_j, C_j > 0$ are finite, and the exponents a, b are common to all coordinates and widths. Then Theorem 2.1 holds, with a sample polynomial depending also on b/a. The prefactors c_j, C_j need not be uniform in width; they affect only the fixed-width small-scale threshold.

Proof. At fixed width there are finitely many coordinates. If s_j denotes $\sqrt{N/d}$, 1, or $N^{-1/2}$ according to layer, then $\kappa_N(w) = \sum_j (w_j/s_j)^2$. A strict target witness obtained by cloning has energy at most B_0 from (10). For every $\zeta > 0$, a sufficiently small positive-volume neighborhood of the witness lies in $U_{N,f}$ and has energy at most $B_0 + \zeta$. Integrating the lower density envelope there gives

$$\liminf_{\sigma \downarrow 0} \sigma^2 \log \mu_{N,\sigma}(U_{N,f}) \ge -bB_0. \tag{14}$$

All powers of σ and fixed density prefactors disappear after multiplication by σ^2.

For any g and $\theta \in (0,1)$, the upper envelope and the inequality $\kappa_N(w) \ge \kappa_N(g)$ on $U_{N,g}$ give

$$\mu_{N,\sigma}(U_{N,g}) \le C_{N,\theta} \exp\{-(1-\theta)a\kappa_N(g)/\sigma^2\}.$$

Here the remaining integral of $\sigma^{-P}\exp\{-\theta a\kappa_N(w)/\sigma^2\}$ is a constant independent of σ. Letting $\theta \downarrow 0$ proves

$$\limsup_{\sigma \downarrow 0} \sigma^2 \log \mu_{N,\sigma}(U_{N,g}) \le -a\kappa_N(g). \tag{15}$$

Set $K = (b/a)B_0 + 1$ and $\mathcal{H} = \{g : \kappa_N(g) \le K\}$. Each function outside $\mathcal{H}$ has mass negligible relative to f by (14)–(15). There are only finitely many Boolean functions, so their total ratio tends to zero. Global-margin conditioning cancels in this ratio. Choose σ_0 so that it is at most $\varepsilon/6$. Lemma 5.1 and Lemma 3.1, with $a = \tau = \varepsilon/6$, give Gibbs risk at most $\varepsilon/3$ as soon as

$$n \ge \frac{6}{\varepsilon}[H_L(d,K) + \log(1/\delta)].$$

The threshold is independent of D, n, δ. Finally use (2). For the curvature family, integrating the bounds on V'' gives (13) with $a = m/2$, $b = M/2$; this proves Theorem 2.1. □

Proposition 5.3 (Unchanged strong-prior ordering)**.** *Suppose the finitely many standardized coordinate densities are positive and continuous, and for a common $a > 0$ satisfy*

$$\lim_{u\to+\infty} \frac{-\log q_j(u)}{u^2} = \lim_{u\to-\infty} \frac{-\log q_j(u)}{u^2} = a.$$

Then for every realizable function g,

$$\lim_{\sigma\downarrow 0} \sigma^2 \log \mu_{N,\sigma}(U_{N,g}) = -a\kappa_N(g).$$

Thus its leading strong-prior ordering is exactly the weighted quadratic energy ordering. That energy is polynomially equivalent to $C_L(g)$ whenever $N \geq C_L(g)$.

Proof. For any $\zeta \in (0, a)$ the tail asymptotics, positivity, and continuity supply global upper and lower envelopes with exponents $a - \zeta$ and $a + \zeta$. Apply the preceding integral estimates to witnesses arbitrarily close to the infimum $\kappa_N(g)$, then let $\zeta \downarrow 0$. Cloning bounds energy polynomially in $d, C_L(g)$; deterministic compression bounds $C_L(g)$ polynomially in $d, \kappa_N(g) + 1$. □

Examples of (13) include Gaussian densities multiplied by factors bounded above and below by positive constants, and centered Gaussian scale mixtures with scales in a fixed compact subinterval of $(0, \infty)$. Neither symmetry nor log-concavity is needed for this strong-prior proposition. The density $\exp\{-u^2/2 - b \log\cosh u\}$ also satisfies Proposition 5.3, because its additional log density is only linear in $|u|$ at infinity. Variance standardization changes the common quadratic coefficient but not the ordering.

6 Finite-scale learning for $\gamma > 1$

Lemma 6.1 (A non-Gaussian singular-value tail)**.** *Let $B = \sigma Z$ be an $N \times N$ matrix with independent ν coordinates. If $a \geq 16\sigma\sqrt{N/m}$ and $B_{>a}$ retains exactly the singular values greater than a, then*

$$\mathbb{P}\{\|B_{>a}\|_F^2 > K\} \leq e^{-mK/(8\sigma^2)}.$$

Proof. The net proof in Section 4 gives

$$\mathbb{P}\{\|B\|_{\mathrm{op}} > 8\sigma\sqrt{N/m}\} \leq 2e^{(2\log 9 - 8)N} < 1/2.$$

The convex ball $\mathcal{B} = \{T : \|T\|_{\mathrm{op}} \leq a/2\}$ therefore has probability at least $1/2$. Its Frobenius distance from B is

$$D(B)^2 = \sum_j (s_j(B) - a/2)_+^2.$$

The enlargement inequality (7), after scaling by σ, implies $\mathbb{P}\{D(B) > r\} \leq e^{-mr^2/(2\sigma^2)}$. Since $s > a$ implies $s \leq 2(s - a/2)$, $\|B_{>a}\|_F \leq 2D(B)$. Take $r = \sqrt{K}/2$. □

For later use, (6) and a one-dimensional Gaussian integration identity give

$$\mathbb{E}e^{m\xi^2/4} \leq \sqrt{2}. \tag{16}$$

Indeed $e^{mx^2/4} = \mathbb{E}_G e^{\sqrt{m/2}Gx}$, with G standard normal; condition first on G. For D_0 independent coordinates, Markov's inequality consequently gives

$$\mathbb{P}\{\textstyle\sum_{j=1}^{D_0} \xi_j^2 > u\} \leq 2^{D_0/2} e^{-mu/4} \leq e^{-mu/8} \quad (mu \geq 8D_0). \tag{17}$$

Lemma 6.2 (Aggregate structural tail)**.** *Set*

$$a_K = [8L(\sqrt{K}+1)^{L-2}]^{-1}, \qquad J_K = (L+3)K, \qquad K \geq 1.$$

If

$$N^{\gamma-1} \geq 8(d+1)/m, \qquad N^{\gamma-1} \geq 256/(ma_K^2), \tag{18}$$

then

$$\mu_{N,\gamma}\left(\bigcup_{g:\kappa_N(g)>J_K} U_{N,g}\right) \leq Le^{-mKN^\gamma/8}.$$

Proof. Consider the event

$$\tfrac{d}{N}\|W^1\|_F^2 \leq K, \quad N\|W^L\|_2^2 \leq K, \quad \|(W^s)_{>a_K}\|_F^2 \leq K \ (2 \leq s < L).$$

The first and last weighted norms are $N^{-\gamma}$ times sums of $N(d+1)$ and N standardized squares. Equation (17) and the first width condition bound their tails by $e^{-mKN^\gamma/8}$. The second width condition permits Lemma 6.1 at each internal layer. A union bound bounds failure of the event by $Le^{-mKN^\gamma/8}$.

For a draw in this event, retain only internal singular values greater than a_K. The retained matrices have operator norm at most $\sqrt{K}$; each discarded matrix has norm at most a_K. If e_s is the Euclidean activation error, then $e_1 = 0$ and $e_s \leq \sqrt{K}e_{s-1} + a_K\sqrt{N}$. The readout norm is at most $\sqrt{K/N}$, so the uniform output error is at most

$$a_K \sum_{j=1}^{L-2} K^{j/2} \leq 1/8.$$

For $L = 2$ there is no change. A strict-margin draw therefore retains margin greater than $3/8$. Doubling its readout restores margin greater than $1/2$ and gives energy at most $K+(L-2)K+4K = J_K$. Every full-margin draw outside the asserted function class must belong to the exceptional parameter event, proving the lemma. □

Proof of Theorem 2.2 for $\gamma > 1$. Write $A = A_L(d, C)$ and choose

$$K = \max\{1, \tfrac{8}{m}[A + \log(6L/\varepsilon)]\}, \qquad T = \left\lceil \max\{C, [8(d+1)/m]^{1/(\gamma-1)}, [256/(ma_K^2)]^{1/(\gamma-1)}\} \right\rceil.$$

For $N \geq T$, divide the aggregate tail by the target lower bound (11). Global-margin normalization cancels and gives

$$\frac{\pi(\{g : \kappa_N(g) > J_K\})}{\pi(f)} \leq Le^{-N^\gamma(mK/8-A)} \leq \varepsilon/6.$$

Necessarily f belongs to the remaining class, since otherwise the ratio would be at least one. Its log cardinality is at most $H_L(d, J_K)$. Lemma 3.1, with $a = \tau = \varepsilon/6$, gives the claimed risk for $n \geq (6/\varepsilon)[H_L(d, J_K) + \log(1/\delta)]$.

For $C \leq N < T$, the target mass gives $-\log \pi(f) \leq AN^\gamma \leq AT^\gamma$. Lemma 3.2 gives the same risk if $n \geq (6/\varepsilon)[AT^\gamma + \log(6/(\varepsilon\delta))]$. At fixed L, γ, m, M, all of K, T, T^γ have polynomial majorants in $d, C, \log(1/\varepsilon)$. Majorize $\max\{H_L(d, J_K), AT^\gamma\}$ by $p_{L,\gamma}$ and use (2). This covers every $N \geq C$. □

7 The mean-field endpoint $\gamma = 1$

At this endpoint an internal non-Gaussian row has the same covariance as its Gaussian counterpart but a different distribution. The following comparison is uniform in the number of input coordinates of the field. The vector dimension below will eventually be 2^d.

Lemma 7.1 (A relative Lindeberg estimate)**.** *Let $a_1, \dots, a_N \in \mathbb{R}^s$ obey $\|a_j\|_\infty \le N^{-1/2}$. Let ξ_j be independent centered variance-one variables with law ν, and let g_j be independent standard Gaussians. Define*

$$Z_\nu = \sum_j a_j \xi_j, \qquad Z_G = \sum_j a_j g_j, \qquad F_\Gamma(z) = \tanh(z)^T \Gamma \tanh(z),$$

where $\|\Gamma\|_{1,\mathrm{ent}} \le G$. Write

$$P(G) = 24G + 36G^2 + 8G^3, \qquad J_\nu = \mathbb{E}[|\xi|^3 e^{|\xi|}] + \mathbb{E}[|g|^3 e^{|g|}], \qquad c_\nu = eJ_\nu/6.$$

If $N \ge 4G^2$, then

$$\left|\log \mathbb{E} e^{F_\Gamma(Z_\nu)} - \log \mathbb{E} e^{F_\Gamma(Z_G)}\right| \le c_\nu P(G)/\sqrt{N}. \tag{19}$$

The constant is finite for the curvature family. The conclusion requires only the displayed moment, centering, and variance assumptions.

Proof. Using $|\tanh|, |\tanh'| \le 1$, $|\tanh''| \le 2$, and $|\tanh'''| \le 6$, differentiation in a direction a with $\|a\|_\infty \le N^{-1/2}$ gives

$$|D_a F_\Gamma| \le 2G/\sqrt{N}, \quad |D_a^2 F_\Gamma| \le 6G/N, \quad |D_a^3 F_\Gamma| \le 24G/N^{3/2}.$$

Thus $|D_a^3 e^{F_\Gamma}| \le P(G) N^{-3/2} e^{F_\Gamma}$. Fix the residual vector z in a one-coordinate replacement and put $f_z(t) = e^{F_\Gamma(z+at)}$ and $\lambda = 2G/\sqrt{N} \le 1$. The first derivative bound implies

$$e^{-\lambda|t|} f_z(0) \le f_z(t) \le e^{\lambda|t|} f_z(0).$$

Taylor expansion through order two with integral remainder yields

$$|f_z(t) - f_z(0) - t f_z'(0) - t^2 f_z''(0)/2| \le \frac{P(G)}{6N^{3/2}} |t|^3 e^{\lambda|t|} f_z(0).$$

The first three averaged terms match for ξ and g. The difference of the one-coordinate expectations is therefore at most $P(G) J_\nu f_z(0)/(6N^{3/2})$. Both expectations are at least $e^{-1} f_z(0)$, since Jensen and variance one give $\mathbb{E} e^{-\lambda|\xi|} \ge e^{-\lambda \mathbb{E}|\xi|} \ge e^{-1}$. The same holds for g.

Averaging over any independent random residual z preserves both bounds. The inequality $|\log u - \log v| \le |u - v|/\min\{u, v\}$ shows that the logarithms differ by at most $c_\nu P(G) N^{-3/2}$ at this replacement. Telescoping over all N coordinates proves (19). No inverse global partition-function bound was used; such a bound would have introduced an unnecessary exponential in G. □

Let $s = 2^d$, $r = L - 1$, and index fields and covariance matrices by all inputs. Write

$$Q_{\ell,N} = \frac{1}{N} \sum_{i=1}^N h_i^\ell (h_i^\ell)^T.$$

For a law μ on $[-1, 1]^s$, define

$$\mathcal{J}_G(\widehat{Q}; \mu) = \max_{\|\Gamma\|_{1,\mathrm{ent}} \le G} \left\{ \langle \Gamma, \widehat{Q} \rangle - \log \mathbb{E}_{h \sim \mu} e^{h^T \Gamma h} \right\}. \tag{20}$$

Appendix B proves, for any law μ, that an empirical second moment from N independent rows satisfies

$$\mathbb{P}\{\mathcal{J}_G(\widehat{Q};\mu) > K\} \le \exp\{q\log(2s^2+1) - N(K-1)\}, \qquad q = \lceil G^2/2 \rceil. \tag{21}$$

It also proves that $\mathcal{J}_G \le K$ supplies a quadratic exponential tilt, of parameter norm at most G, whose second moment is within K/G of $\widehat{Q}$ in entrywise maximum norm. These facts are distribution-free.

Let μ_0 be the actual first-layer activation-field law, obtained from independent ν weights divided by $\sqrt{d}$. Keep this law exactly; its dimension need not tend to infinity. For later layers write M_Q for the activation law of a Gaussian field of covariance Q, and $\mathsf{F}_\Gamma(Q)$ for its second moment after tilting by $e^{h^T\Gamma h}$. Lemma C.1 proves

$$\|\mathsf{F}_\Gamma(Q) - \mathsf{F}_\Gamma(Q')\|_{\max} \le 8(1+G)^2\|Q-Q'\|_{\max}. \tag{22}$$

This comparison concerns an auxiliary Gaussian family; the Bayesian learner still has non-Gaussian weights at every layer.

Conditional on the preceding activations $h_1,\dots,h_N$, an internal row has preactivation $N^{-1/2}\sum_j \xi_j h_j$. Apply Lemma 7.1 with $a_j = h_j/\sqrt{N}$. Its Gaussian comparison has covariance exactly $Q = N^{-1}\sum_j h_j h_j^T$. If μ_{act} is the true conditional activation law, then

$$\mathcal{J}_G(\widehat{Q};\mathsf{M}_Q) \le \mathcal{J}_G(\widehat{Q};\mu_{\mathrm{act}}) + c_\nu P(G)/\sqrt{N}. \tag{23}$$

This step does not assume that the actual row law is determined by Q.

7.1 A likely finite class

Let $K \ge 4$ and $\bar{K} = K+1$. Choose the budgets backward:

$$G_r = 32r\bar{K}^2, \qquad G_\ell = 8G_{\ell+1}(1+G_{\ell+1})^2 \quad (\ell < r), \qquad b_\ell = 8(1+G_\ell)^2. \tag{24}$$

Set $q_1 = \lceil G_1^2/2 \rceil$, $\ell_0 = \log(2s^2+1)$, and

$$T(K) = \left\lceil \max\left\{1, \frac{2q_1\ell_0}{K-1}, 4G_1^2, c_\nu^2 P(G_1)^2\right\}\right\rceil. \tag{25}$$

These quantities have polynomial majorants in d, K at each fixed depth: $\ell_0 \le (2d+2)\log 2$ and the recursion in (24) has only the fixed number r of steps.

Lemma 7.2 (Likely proxy covariances)**.** *There is a constant $c > 0$, depending only on the coordinate law, such that for all sufficiently large fixed-law K and $N \ge T(K)$, except on a parameter event of probability at most Le^{-cNK}, the following hold simultaneously: $N\|W^L\|_2^2 \le K$, and there are symmetric matrices Γ_ℓ with $\|\Gamma_\ell\|_{1,\mathrm{ent}} \le G_\ell$ such that the tower*

$$Q_1^* = \frac{\mathbb{E}_{h\sim\mu_0}[hh^T e^{h^T\Gamma_1 h}]}{\mathbb{E}_{h\sim\mu_0} e^{h^T\Gamma_1 h}}, \qquad Q_\ell^* = \mathsf{F}_{\Gamma_\ell}(Q_{\ell-1}^*) \ (\ell \ge 2)$$

satisfies $\|Q_{r,N} - Q_r^*\|_{\max} \le 1/(32\bar{K})$.

Proof. Apply (21) to each actual conditional row law, including μ_0 in the first layer. The threshold gives each failure probability at most $e^{-N(K-1)/2} \le e^{-NK/4}$. By (23) and (25), every internal Gaussian certificate is at most $K+1 = \bar{K}$. The first-layer certificate is at most K and needs no comparison. The bounded-dual lift in Appendix B therefore supplies the tilt matrices with local covariance error at most $\bar{K}/G_\ell$.

Let $e_\ell = \|Q_{\ell,N} - Q^*_\ell\|_{\max}$. Then $e_1 \le \bar{K}/G_1$, and $e_\ell \le \bar{K}/G_\ell + b_\ell e_{\ell-1}$ for $\ell \ge 2$, by (22). The backward budgets give

$$e_r \le \sum_{\ell=1}^{r} \frac{\bar{K}}{G_\ell} \prod_{j=\ell+1}^{r} b_j = \frac{1}{32\bar{K}}.$$

For the readout, (16) gives

$$\mathbb{P}\{N\|W^L\|_2^2 > K\} \le \exp\{-mNK/4 + (N/2)\log 2\} \le e^{-mNK/8}$$

when $K \ge 4\log 2/m$. Thus $c = \min\{1/4, m/8\}$ works after requiring $K \ge \max\{4, 4\log 2/m\}$. □

Lemma 7.3 (A bounded dual readout)**.** *Suppose H is an $s \times N$ feature matrix and a readout v obeys $g_i(Hv)_i > 1/2$ and $N\|v\|_2^2 \le K$. Put $Q = HH^T/N$. There is $\beta \in \mathbb{R}^s$ with $\|\beta\|_1 \le 2K$ and $g_i(Q\beta)_i \ge 1/2$ for every i.*

Proof. Minimize $N\|u\|_2^2/2$ subject to $g_i(Hu)_i \ge 1/2$. The strictly feasible v gives the convex optimality conditions with multipliers $\lambda_i \ge 0$: $Nu = H^T\beta$, $\beta_i = g_i\lambda_i$, and $\lambda_i[g_i(Hu)_i - 1/2] = 0$. Hence $Hu = Q\beta$ and

$$\tfrac{1}{2}\|\beta\|_1 = \beta^T Hu = N\|u\|_2^2 \le K.$$

No inversion or nonsingularity of Q is needed. □

Proposition 7.4 (Mean-field aggregate mass bound)**.** *For each K as in Lemma 7.2, there is a finite class $\mathcal{H}_K$, independent of learner width, with $\log|\mathcal{H}_K| \le \widetilde{H}_{L,\nu}(d,K)$ for a polynomially bounded function at fixed depth, such that*

$$\mu_{N,1}\left(\bigcup_{g\notin\mathcal{H}_K} U_{N,g}\right) \le Le^{-cNK} \qquad (N \ge T(K)).$$

Proof. Fix a full-margin draw in the likely set of Lemma 7.2. Lemma 7.3 gives $\|\beta\|_1 \le 2K$ and margin at least $1/2$ for $Q_{r,N}\beta$. Replacing $Q_{r,N}$ by Q^*_r changes the output by at most $2K/(32\bar{K}) \le 1/16$. Consequently $2Q^*_r\beta$ has signed margin at least $7/8$.

Proposition C.2 realizes this tower within $1/8$ by a finite depth-L network with width and weights polynomially bounded in $d, \bar{K}, G_1$. Its first hidden layer is sampled from the true first-layer law with the tilt; subsequent layers use Gaussian rows to realize the auxiliary Gaussian tower. This constructs a deterministic short description of the sign function, without changing the learner. The realized margin is at least $3/4$. Define $\mathcal{H}_K$ to be all sign functions admitting these common width and weight bounds with margin at least $3/4$, padding by zero neurons to the common width when needed. Lemma C.3 bounds this class's logarithmic cardinality polynomially in d, K at fixed depth. In particular every member of $\mathcal{H}_K$ has the stated bounded implementation; the grid is used only to count such functions. Every full-margin draw outside this class must fail the likely event, giving the stated aggregate bound. □

Proof of Theorem 2.2 for $\gamma = 1$. Let $A = A_L(d, C)$ and choose

$$K \ge \max\{4, 4\log 2/m,\ c^{-1}[A + \log(6L/\varepsilon)]\}.$$

For $N \ge T(K)$, Proposition 7.4 and the target mass give

$$\frac{\pi(\mathcal{H}_K^c)}{\pi(f)} \le Le^{-N(cK-A)} \le \varepsilon/6.$$

The target therefore belongs to $\mathcal{H}_K$. Lemma 3.1 gives risk at most $\varepsilon/3$ provided $n \geq (6/\varepsilon)[\widetilde{H}_{L,\nu}(d,K)+\log(1/\delta)]$. For every remaining width $C \leq N < T(K)$, the target lower bound gives $-\log \pi(f) \leq AT(K)$. Lemma 3.2 supplies the same risk at sample size $(6/\varepsilon)[AT(K) + \log(6/(\varepsilon\delta))]$. Both $\widetilde{H}_{L,\nu}(d,K)$ and $AT(K)$ have polynomial majorants in $d, C, \log(1/\varepsilon)$ at fixed depth. Taking a common majorant and applying (2) completes the proof. □

8 Prior mass and circuit complexity

The preceding arguments also compare function probability with bounded-weight implementation complexity. This is stronger than merely having a sample bound, although it does not specify the relative order of functions whose complexities are close.

Corollary 8.1 (A polynomial prior-complexity comparison)**.** *Fix L, $\gamma \geq 1$, and a coordinate law in the main family. For a function with $\mu_{N,\gamma}(U_{N,f}) > 0$, define its raw mass cost*

$$b_{N,\gamma}(f) = -N^{-\gamma} \log \mu_{N,\gamma}(U_{N,f}).$$

There is a polynomial $P_{L,\gamma}$, independent of width, such that

$$C_L(f) \leq P_{L,\gamma}(d, 1 + b_{N,\gamma}(f)).$$

Conversely, whenever $N \geq C_L(f)$,

$$b_{N,\gamma}(f) \leq A_L(d, C_L(f)).$$

The two functions of complexity on the right have polynomial majorants at fixed depth and exponent. Global-margin normalization is common to all functions, so using raw masses does not change their ordering at a fixed width and exponent.

Proof. The converse is Proposition 4.3. Put $b = b_{N,\gamma}(f) \geq 0$. If $\gamma > 1$, choose $K \geq \max\{1, (8/m)[b+\log(2L)]\}$. Above the width threshold in (18), Lemma 6.2 makes the total mass outside the bounded-energy class at most

$$Le^{-[b+\log(2L)]N^\gamma} < e^{-bN^\gamma} = \mu_{N,\gamma}(U_{N,f}).$$

Thus $U_{N,f}$ contains a good draw and $\kappa_N(f) \leq J_K$. Deterministic compression bounds $C_L(f)$ polynomially in $d, 1+b$. The threshold itself is polynomially bounded in these variables for fixed L, γ, m.

At $\gamma = 1$, choose K above the fixed-law thresholds with $cK \geq b + \log(2L)$. For $N \geq T(K)$, the same mass comparison in Proposition 7.4 implies $f \in \mathcal{H}_K$. The constructive proof of that proposition gives a network of width and weights polynomially bounded in d, K. Padding by zero neurons if necessary shows that $C_L(f)$ has the required bound.

It remains to handle widths below the respective polynomial thresholds. Let $D_N = N(d+2) + (L-2)N^2$ be the number of scalar parameters, and let S be the sum of their standardized squares. Exactly, $S = N^\gamma \kappa_N(w)$. By the first inequality in (17), at

$$u = \frac{4bN^\gamma + 2D_N \log 2 + 4}{m}$$

one has $\mathbb{P}\{S > u\} \leq e^{-bN^\gamma - 1}$. This is strictly less than the mass of $U_{N,f}$, so that region contains a point with

$$\kappa_N(w) \leq \frac{u}{N^\gamma} \leq \frac{4b + 2D_N \log 2 + 4}{m}.$$

Since N is below a polynomial threshold in $d, 1+b$, this is a polynomial energy bound. Deterministic compression proves the assertion in the remaining width range. □

This comparison concerns the stated bounded-weight neural-network complexity at fixed depth. It does not identify prior probability with an exponential of ordinary Boolean circuit description length, and it does not assert that posterior inference returns a minimum circuit. The exact strong-prior rate under quadratic recession, rather than merely its polynomial comparison, is Proposition 5.3.

9 Scope and remaining questions

The assumptions identify a broad sufficient family rather than a classification of non-Gaussian priors. In particular, Laplace, Student, uniform, and pure quartic-exponential coordinate laws are not covered by Theorem 2.2. Strong log-concavity alone would allow superquadratic tails, whereas the upper curvature bound here prices a translated target quadratically. A compactly supported coordinate law has a more immediate difficulty in the fixed-width strong-prior limit: for sufficiently small scale all possible outputs may fail the global margin, leaving the conditioned learner undefined.

The strong-prior theorem actually needs only the two-sided Gaussian envelopes (13) and permits asymmetric and non-log-concave densities; see Section 5. At finite scale, symmetry supplies centering in the translated conditional law, while strong log-concavity controls that law even after a convex cylinder restriction. Removing either condition requires a replacement for that target-mass argument. The mean-field comparison itself needs substantially less: centered variance-one internal coordinates with $\mathbb{E}[|\xi|^3 e^{|\xi|}] < \infty$, suitable square-exponential moments at the boundary layers, and a separate target-mass lower bound.

The theorems preserve the original global margin restriction. They do not assert the same result after conditioning only on margins at training points, under label noise, or for ReLU activations. Neither do they settle which non-Gaussian laws at their natural width scales retain exactly the same ordering of functions by prior probability, beyond the polynomial comparison proved in Corollary 8.1. For laws with asymptotically quadratic log density, the strong-prior ordering is exactly the same weighted quadratic-energy ordering; see Proposition 5.3.

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A Compression of bounded-energy networks

The following deterministic result supplies the width-independent count used in the strong-prior and $\gamma > 1$ arguments. It depends on neither Gaussianity nor any probabilistic property of the weights.

Theorem A.1 (Compression)**.** *Let $d \geq 1$, $L \geq 2$, and let $\phi : \mathbb{R} \to [-1, 1]$ be 1-Lipschitz. Suppose a width-N network W satisfies $\kappa_N(W) \leq K$, where $K \geq 1$. Set*

$$\begin{aligned} \eta &= \frac{1}{12(L-1)}, \qquad \rho = \eta 2^{-L} K^{-(L-1)/2}, \\ m_L(d, K) := M &= \max\left\{ \lceil 9K \rceil, \left\lceil 256\rho^{-4}\big(d + \log(L/\rho)\big) \right\rceil \right\}. \end{aligned} \tag{26}$$

There is a width-M network $\widetilde{W}$ such that

$$\max_{x \in \{-1,1\}^d} |z_{\widetilde{W}}(x) - z_W(x)| \leq \frac{1}{12}, \qquad \kappa_M(\widetilde{W}) \leq 4K. \tag{27}$$

If W implements a Boolean function g with strict margin $1/2$, then multiplying the readout of $\widetilde{W}$ by $3/2$ gives a width-M implementation of g with strict margin $1/2$, weighted norm at most $9K$, and every weight of absolute value at most M. In particular, $C_L(g) \leq M$. For fixed L, M is bounded by a polynomial in d and K.

Proof. We first truncate large entries and then sample the hidden neurons. All norms of rows below are Euclidean, and all matrix norms with subscript F are Frobenius norms.

Truncation. For $2 \leq s \leq L$, let $r_i^s = \|W_{i,*}^s\|_2$. In each nonzero row, delete entries larger in absolute value than

$$\tau_i^s = \frac{2r_i^s}{\rho\sqrt{N}}.$$

Leave zero rows unchanged, write $\overline{W}^s$ for the resulting matrices, and set $\overline{W}^1 = W^1$. Here and below W^L is the single readout row. Because all hidden activations have absolute value at most one, deleting a row's large entries changes its preactivation by at most

$$\sum_{j:\, |W_{ij}^s| > \tau_i^s} |W_{ij}^s| \leq \frac{(r_i^s)^2}{\tau_i^s} = \frac{\rho}{2}\sqrt{N}\, r_i^s.$$

The vector of changes at an internal layer therefore has norm at most $(\rho/2)\sqrt{N}\|W^s\|_F$. Every original or truncated internal matrix has operator norm at most $\sqrt{K}$, and the readout has norm at most $\sqrt{K/N}$. The activation is a contraction in Euclidean norm. Thus truncating layer $s < L$ changes the output by at most

$$\frac{\rho}{2}\sqrt{NK}\, K^{(L-s-1)/2}\sqrt{K/N} = \frac{\rho}{2}K^{(L-s+1)/2} \leq \frac{\eta}{2}.$$

Truncating the readout changes it by at most $(\rho/2)\sqrt{K} \leq \eta/2$. These bounds also hold when other layers have already been truncated. Telescoping over the $L-1$ truncations gives total output error at most $1/24$.

Independent samples and three events. For each hidden layer s, independently choose a list $I_s(1), \ldots, I_s(M)$ of independent uniform indices in $\{1, \ldots, N\}$. Repeated indices represent separate copies of a neuron. Define the fully sampled network by

$$\begin{aligned} \widetilde{W}_{a,*}^1 &= W_{I_1(a),*}^1, \\ \widetilde{W}_{a,b}^s &= \frac{N}{M}\overline{W}_{I_s(a),I_{s-1}(b)}^s \qquad\qquad (2 \leq s \leq L-1), \\ \widetilde{W}_b^L &= \frac{N}{M}\overline{W}_{I_{L-1}(b)}^L. \end{aligned}$$

For the analysis expose these independent lists in reverse order, $I_{L-1}, \dots, I_1$. At the stage sampling layer m, all earlier layers still have their original, truncated weights. Let $\overline{h}_j^m(x)$ denote their exactly computed activations. The estimated sum into a retained row i of layer $m+1$ is

$$\frac{N}{M} \sum_{b=1}^{M} \overline{W}_{i,I_m(b)}^{m+1} \overline{h}_{I_m(b)}^m(x).$$

Its expectation is the corresponding full sum over N neurons. Conditional on the later lists, its summands are independent and bounded in absolute value by $(N/M)\tau_i^{m+1}$. Hoeffding's inequality [1] shows that its deviation exceeds $(\rho/2)\sqrt{N} r_i^{m+1}$ with probability at most

$$2 \exp\left(-\frac{\rho^4 M}{32}\right).$$

A zero row is estimated exactly. The bound is uniform in the retained row, so it remains valid after averaging over the later lists.

Let A be the event that all these row estimates, at all stages and all 2^d inputs, satisfy their deviation bounds. At the last hidden layer there is one output row; at every other stage there are M retained rows. Consequently

$$\mathbb{P}(A^c) \le 2^{d+1} LM \exp(-\rho^4 M/32) \le \frac{1}{100}. \tag{28}$$

For completeness, put $a = d + \log L + \log(1/\rho)$ and $M_0 = \lceil 256 \rho^{-4} a \rceil$. Then $M_0 \le 257 \rho^{-4} a$, and $u \exp(-\rho^4 u/32)$ decreases for $u \ge M_0$. Substituting M_0 and using $\log a \le a - 1$ bounds the logarithm of the right side before its final inequality by

$$-(7 - \log 2)d - 6 \log L - 3 \log(1/\rho) + \log 514 - 1 \le -\log 100.$$

This also proves the bound when M was enlarged to be at least $9K$.

Next define the total retained row mass using the *original* row norms:

$$R = N \sum_{s=2}^{L-1} \sum_{a=1}^{M} (r_{I_s(a)}^s)^2 + N \|W^L\|_2^2.$$

Its expectation is

$$\mathbb{E}R = M \sum_{s=2}^{L-1} \|W^s\|_F^2 + N \|W^L\|_2^2 \le MK.$$

By Markov's inequality the event $B = \{R \le 8MK\}$ has probability at least $7/8$. In particular, on B the retained row mass at each individual stage is at most $8MK$.

Finally, direct calculation gives

$$\begin{aligned} \mathbb{E}\left[\frac{d}{M} \|\widetilde{W}^1\|_F^2\right] &= \frac{d}{N} \|W^1\|_F^2, \\ \mathbb{E}\|\widetilde{W}^s\|_F^2 &= \|\overline{W}^s\|_F^2 \qquad (2 \le s \le L-1), \\ \mathbb{E}\left[M \|\widetilde{W}^L\|_2^2\right] &= N \|\overline{W}^L\|_2^2. \end{aligned}$$

For an internal matrix, independence of the two adjacent layers' index lists makes each index pair uniform on $\{1, \dots, N\}^2$; the factor $(N/M)^2$ is canceled by the M^2/N^2 sampling factor. Independence

among different matrix entries is unnecessary. Thus $\mathbb{E}\kappa_M(\widetilde{W}) \le K$, and the event $C = \{\kappa_M(\widetilde{W}) \le 4K\}$ has probability at least $3/4$. On C every sampled internal matrix has operator norm at most $2\sqrt{K}$, and the sampled readout has norm at most $2\sqrt{K/M}$.

The events need not be independent:

$$\mathbb{P}(A \cap B \cap C) \ge 1 - \frac{1}{100} - \frac{1}{8} - \frac{1}{4} > 0.$$

Fix a collection of lists in their intersection.

Reverse-order telescoping. Let F_m be the network with hidden layers $m, \dots, L-1$ sampled and layers $1, \dots, m-1$ still original and truncated. Incoming weights to a sampled layer select its rows, and outgoing weights from a sampled layer are multiplied by N/M. This defines the intermediate networks with different widths in different layers. In particular, F_L is the truncated network and F_1 is the fully sampled network.

When comparing F_{m+1} and F_m, layer m computes the original truncated neuron functions before selection. The only new discrepancy is in the sums into the retained neurons of layer $m+1$. For $m = L-1$, event A bounds the output discrepancy directly by $(\rho/2)\sqrt{K} \le \eta/2$. For $m < L-1$, put $t = L-m$. Events A and B bound the Euclidean discrepancy in layer $m+1$'s preactivation by $(\rho/2)\sqrt{8MK}$. All matrices *after* layer $m+1$ already equal their final sampled matrices, so their operator norms are controlled by C. Propagating the discrepancy therefore changes the output by at most

$$\begin{aligned}\frac{\rho}{2}\sqrt{8MK}\,(2\sqrt{K})^{t-2}2\sqrt{K/M} &= \rho\, 2^{t-1/2}K^{t/2} \\ &\le \eta\, 2^{-3/2} \le \frac{\eta}{2}, \qquad 2 \le t \le L-1.\end{aligned}$$

There are $L-1$ sampling stages, contributing at most $1/24$ in total. Adding the truncation error proves the approximation bound in (27); event C proves its norm bound.

If the original network has strict margin $1/2$, the sampled network has strict margin $5/12$. Multiplying its linear readout by $3/2$ restores strict margin $1/2$ and multiplies the weighted norm by at most $9/4$. At width M the resulting entrywise bounds are

$$|W^1_{aj}| \le \sqrt{9KM/d}, \qquad |W^s_{ab}| \le 3\sqrt{K}\ (1 < s < L), \qquad |W^L_b| \le 3\sqrt{K/M}.$$

Each is at most M because $d \ge 1$ and $M \ge 9K \ge 9$. The definition of C_L now applies. Finally, substituting (26) gives $\rho^{-4} = 12^4(L-1)^4 16^L K^{2(L-1)}$, which establishes the stated polynomial dependence at fixed depth. □

B A finite family of entropy certificates

The following bound replaces a discretization of the full feature space by sparse linear tests. Its logarithmic dependence on the number of observables is essential when those observables are indexed by all pairs of Boolean inputs.

Let $(\Omega, \mathcal{A}, \mu)$ be any probability space, let $F = (F_1, \dots, F_D) : \Omega \to [-1,1]^D$ be measurable, and write

$$\Lambda(\lambda) = \log \int e^{\lambda \cdot F}\, d\mu, \qquad \Psi_G(q) = \max_{\|\lambda\|_1 \le G} \{\lambda \cdot q - \Lambda(\lambda)\}, \qquad G > 0.$$

The maximum exists: its domain is compact and its objective is continuous. The norm $\|\lambda\|_1$ here is the sum of the absolute values of its D coordinates.

Lemma B.1 (A bounded exponential tilt with controlled error)**.** *For every $q \in \mathbb{R}^D$, there is a maximizer λ_* in the preceding display such that the probability measure*

$$\frac{d\nu_*}{d\mu} = \exp\{\lambda_* \cdot F - \Lambda(\lambda_*)\}$$

satisfies the exact identity

$$\Psi_G(q) = D(\nu_* \| \mu) + G \left\| q - \int F \, d\nu_* \right\|_\infty . \tag{29}$$

Consequently, if $\Psi_G(q) \le K$, then

$$D(\nu_* \| \mu) \le K, \qquad \left\| q - \int F \, d\nu_* \right\|_\infty \le K/G.$$

Proof. Boundedness of F permits differentiation under the integral and gives $\nabla\Lambda(\lambda_*) = \int F \, d\nu_* =: m_*$. The first-order condition for a differentiable concave function maximized over the convex ℓ^1 ball is

$$(q - m_*) \cdot (\lambda - \lambda_*) \le 0 \qquad (\|\lambda\|_1 \le G).$$

Taking the maximum of the left linear functional over that ball gives

$$\lambda_* \cdot (q - m_*) = G\|q - m_*\|_\infty.$$

Moreover, $D(\nu_* \| \mu) = \lambda_* \cdot m_* - \Lambda(\lambda_*)$. Adding these two identities proves (29). Relative entropy is nonnegative, which proves the stated consequences. The log density of ν_* relative to μ is bounded, so its entropy is finite. No support, density, or nonsingularity assumption on μ was used. □

For an integer $s \ge 1$, introduce the finite set

$$\mathcal{V}_{G,s} = \left\{ \frac{1}{s} \sum_{j=1}^{s} a_j : a_j \in \{0, Ge_1, -Ge_1, \ldots, Ge_D, -Ge_D\} \right\}.$$

Thus $\#\mathcal{V}_{G,s} \le (2D+1)^s$ and every vector in this set has ℓ^1 norm at most G.

Lemma B.2 (Sparse approximation of the dual value)**.** *For every $q \in \mathbb{R}^D$,*

$$\Psi_G(q) \le \frac{G^2}{2s} + \max_{v \in \mathcal{V}_{G,s}} \{v \cdot q - \Lambda(v)\}. \tag{30}$$

Proof. Fix λ with $\|\lambda\|_1 \le G$. Let the random vector A take the value $G \operatorname{sign}(\lambda_i) e_i$ with probability $|\lambda_i|/G$ for each nonzero coordinate, and take value zero with the remaining probability. Then $\mathbb{E}A = \lambda$. Take independent copies $A_1, \ldots, A_s$ and put $V = s^{-1} \sum_j A_j \in \mathcal{V}_{G,s}$. For each fixed $\omega \in \Omega$, the scalar $A \cdot F(\omega)$ lies in $[-G, G]$. Hoeffding's exponential-moment inequality therefore gives

$$\mathbb{E} \exp\{(V - \lambda) \cdot F(\omega)\} \le \exp\{G^2/(2s)\}.$$

Tonelli's theorem and Jensen's inequality imply

$$\begin{aligned} \mathbb{E}\Lambda(V) &= \mathbb{E} \log \int e^{V \cdot F} \, d\mu \\ &\le \log \mathbb{E} \int e^{V \cdot F} \, d\mu \\ &\le \Lambda(\lambda) + G^2/(2s). \end{aligned}$$

Since $\mathbb{E}(V \cdot q) = \lambda \cdot q$, at least one realization $v \in \mathcal{V}_{G,s}$ satisfies

$$v \cdot q - \Lambda(v) \geq \lambda \cdot q - \Lambda(\lambda) - G^2/(2s).$$

Take λ to be a maximizer defining $\Psi_G(q)$. □

Proposition B.3 (A nonasymptotic entropy certificate)**.** *Let $X_1, \dots, X_N$ be independent with law μ, and put $q_N = N^{-1} \sum_{i=1}^N F(X_i)$. For every $K > 0$, $G > 0$, and integer $s \geq 1$,*

$$\mathbb{P}\{\Psi_G(q_N) > K\} \leq \exp\left\{ s \log(2D+1) - N\left(K - \frac{G^2}{2s}\right)\right\}. \tag{31}$$

Outside this exceptional event, the empirical moment vector is within K/G in maximum norm of the moments of an exponential tilt with parameter ℓ^1 norm at most G and relative entropy at most K. In particular, choosing $s = \lceil G^2 \rceil$ bounds the right side by $\exp\{\lceil G^2 \rceil \log(2D+1) - N(K - 1/2)\}$.

Proof. For each fixed $v \in \mathcal{V}_{G,s}$, independence gives exactly

$$\mathbb{E} \exp\{N[v \cdot q_N - \Lambda(v)]\} = 1.$$

Markov's inequality bounds the probability that its bracket exceeds $K - G^2/(2s)$ by $\exp\{-N(K - G^2/(2s))\}$. The sparse approximation lemma and a union bound over $\mathcal{V}_{G,s}$ prove (31). The bounded-dual lift lemma gives the final assertion. □

Application to hidden-layer covariances. Take m input points, $D = m^2$, $\mu = \mathcal{N}(0, Q)$ for any positive semidefinite covariance Q, and $F_{ij}(z) = \tanh(z_i)\tanh(z_j)$, indexed by all ordered pairs (i, j). Reshape a maximizing parameter λ_* into a matrix and replace it by its symmetric part Γ. This preserves its quadratic form and does not increase its entrywise ℓ^1 norm. The lifted covariance is then exactly

$$\mathsf{F}_\Gamma(Q) = \frac{\mathbb{E}_{Z \sim \mathcal{N}(0,Q)}[\tanh(Z)\tanh(Z)^\top e^{\tanh(Z)^\top \Gamma \tanh(Z)}]}{\mathbb{E}_{Z \sim \mathcal{N}(0,Q)} e^{\tanh(Z)^\top \Gamma \tanh(Z)}}.$$

Thus, except with the probability in (31), the empirical activation covariance $\widehat{Q}$ satisfies

$$\|\widehat{Q} - \mathsf{F}_\Gamma(Q)\|_{\max} \leq K/G, \qquad \|\Gamma\|_{1,\mathrm{ent}} \leq G.$$

This assertion applies directly to Gaussian conditional row laws, uniformly in their covariance, including singular covariances. For the actual non-Gaussian rows in this paper, apply the distribution-free certificate to their true law and then use the relative Lindeberg comparison (23). On the Boolean cube, $m = 2^d$ and the logarithmic factor is $\log(2m^2 + 1) \leq (2d + 2)\log 2$, which is linear in d.

C Covariance stability and finite realization

Write $\phi = \tanh$. Write $\|A\|_{\max} = \max_{ij} |A_{ij}|$ and $\|A\|_{1,\mathrm{ent}} = \sum_{ij} |A_{ij}|$ for matrices on the finite input set. These are different norms; their dual pairing is what avoids a factor of 2^d below. For a positive semidefinite covariance Q and a symmetric matrix Γ, let

$$\mathsf{F}_\Gamma(Q) = \frac{\mathbb{E}_{Z \sim \mathcal{N}(0,Q)}[\phi(Z)\phi(Z)^\top \exp\{\phi(Z)^\top \Gamma \phi(Z)\}]}{\mathbb{E}_{Z \sim \mathcal{N}(0,Q)}[\exp\{\phi(Z)^\top \Gamma \phi(Z)\}]}.$$

All activations are coordinatewise. The definition is valid for singular Q.

Lemma C.1 (A dimension-independent covariance comparison)**.** *If $g = \|\Gamma\|_{1,\mathrm{ent}}$, then for all positive semidefinite Q, Q' of the same size,*

$$\|\mathsf{F}_\Gamma(Q) - \mathsf{F}_\Gamma(Q')\|_{\max} \le 8(1+g)^2 \|Q - Q'\|_{\max}.$$

Proof. Put $V(z) = \phi(z)^\top \Gamma \phi(z)$ and $Q_t = (1-t)Q + tQ'$. For a twice differentiable bounded test function F, let $\mathbb{E}_t$ denote expectation under the Gaussian of covariance Q_t reweighted by e^V. Gaussian differentiation and the quotient rule give

$$\frac{d}{dt}\mathbb{E}_t F = \frac{1}{2}\sum_{a,b}(Q'-Q)_{ab}\{\mathbb{E}_t[\partial_{ab}F + \partial_a F \partial_b V + \partial_b F \partial_a V] + \mathrm{Cov}_t(F, \partial_{ab}V + \partial_a V \partial_b V)\}.$$

This formula has no reciprocal partition-function bound: all terms are expectations under a probability measure. Define $M_1(H) = \sup_z \sum_a |\partial_a H(z)|$ and $M_2(H) = \sup_z \sum_{a,b} |\partial_{ab} H(z)|$. Because $|\phi|, |\phi'| \le 1$ and $|\phi''| \le 2$,

$$M_1(V) \le 2g, \qquad M_2(V) \le 6g.$$

For $F(z) = \phi(z_i)\phi(z_j)$, including $i = j$, $\|F\|_\infty \le 1$, $M_1(F) \le 2$, and $M_2(F) \le 6$. The absolute derivative is therefore at most

$$\frac{1}{2}\|Q - Q'\|_{\max}\{6 + 8g + 2(6g + 4g^2)\} = (3 + 10g + 4g^2)\|Q - Q'\|_{\max}.$$

Integrating in t proves the assertion, with the displayed slightly larger constant. One may initially replace both covariances by $Q + \epsilon I$ and $Q' + \epsilon I$. The functions and all derivatives used above are bounded, and the partition functions lie between e^{-g} and e^g, so weak Gaussian convergence lets $\epsilon \downarrow 0$. This also proves the singular case. □

Proposition C.2 (Finite realization of a bounded-tilt tower)**.** *Fix $L \ge 2$, $K \ge 1$, and $G \ge 1$. Let μ_0 be the actual first-layer activation-field law, with independent incoming coordinates $\xi_j/\sqrt{d}$ of law ν. Suppose symmetric matrices $\Gamma_1, \dots, \Gamma_{L-1}$ and a vector $\beta \in \mathbb{R}^{\mathcal{X}}$ satisfy*

$$\|\Gamma_s\|_{1,\mathrm{ent}} \le G, \qquad \|\beta\|_1 \le 4K.$$

Define Q_1 by tilting μ_0 with $e^{h^T \Gamma_1 h}$, and $Q_s = \mathsf{F}_{\Gamma_s}(Q_{s-1})$ for $s \ge 2$. Then the function $z(x) = (Q_{L-1}\beta)_x$ can be uniformly approximated to accuracy $1/8$ by a depth-L network whose width and weights are bounded by $\mathrm{poly}_L(d, K, G)$. If $f(x)z(x) \ge 7/8$ everywhere, the finite network implements f with margin at least $3/4$.

Proof. Let $b = 8(1+G)^2$ and

$$\eta = [32K(L-1)b^{L-2}]^{-1}.$$

Choose the common width n to be the least integer satisfying

$$n \ge 2\eta^{-2}\log(8(L-1)\,2^{2d}).$$

The construction is random only to prove existence. The matrices Γ_s may be indexed by the entire cube; they select the sampling laws, but are not stored or evaluated by the resulting finite network. No efficient sampling or construction algorithm is claimed. At the first layer, independently sample n weight vectors from their actual independent ν coordinate law divided by $\sqrt{d}$, tilted by

$$\exp\{h(w_i)^\top \Gamma_1 h(w_i)\}, \qquad h(w_i)_x = \phi(w_i \cdot \bar{x}).$$

Their activation vectors have exactly the tilted law defining Q_1. At each subsequent hidden layer, conditional on the already sampled previous activation vectors $h_1, \dots, h_n$, independently sample n standard Gaussian row vectors $\xi_j \in \mathbb{R}^n$ tilted by

$$\exp\{\phi(Z_j)^\top \Gamma_s \phi(Z_j)\}, \qquad Z_j(x) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \xi_{ji} h_i(x).$$

Use the actual network weights $\xi_{ji}/\sqrt{n}$. If $\widehat{Q}_{s-1} = n^{-1} \sum_i h_i h_i^\top$ is the preceding empirical activation covariance, then, conditionally, Z_j before tilting is Gaussian with exactly that covariance. Thus the new activation vectors are conditionally independent, with second moment $\mathsf{F}_{\Gamma_s}(\widehat{Q}_{s-1})$.

Every covariance entry is the empirical mean of n variables in $[-1, 1]$. Hoeffding's inequality, a union bound over 2^{2d} entries, and conditioning at successive layers show, with probability at least $3/4$, that

$$\|\widehat{Q}_1 - Q_1\|_{\max} \le \eta, \qquad \|\widehat{Q}_s - \mathsf{F}_{\Gamma_s}(\widehat{Q}_{s-1})\|_{\max} \le \eta \quad (2 \le s < L).$$

Lemma C.1 then implies

$$\|\widehat{Q}_{L-1} - Q_{L-1}\|_{\max} \le (L-1) b^{L-2} \eta.$$

For each last-layer activation vector h_i, choose the deterministic readout weight $v_i = (\beta \cdot h_i)/n$. The resulting output is $\widehat{z}(x) = (\widehat{Q}_{L-1}\beta)_x$, and therefore

$$\|\widehat{z} - z\|_\infty \le 4K(L-1) b^{L-2} \eta = 1/8.$$

Although β is indexed by the entire input cube, its only role in the finite network is to select these n real readout weights; each satisfies $|v_i| \le 4K/n$. It is not an additional computational input to the network.

It remains to bound the sampled weights without losing polynomial size. For every sampled row, its tilted density relative to its reference is at most e^{2G}, since the quadratic tilt lies in $[-G, G]$. Put $a_0 = \min\{m/4, 1/4\}$ and $B_0 = \sqrt{2}$. Equation (16) and the Gaussian square moment show that every raw reference coordinate obeys $\mathbb{E} e^{a_0 \xi^2} \le B_0$. This also bounds the actual first-layer coordinates divided by $\sqrt{d}$. Let $D_n = n(d+1) + (L-2)n^2$ and choose

$$B = \left\lceil \frac{2G + \log(4B_0 \max\{D_n, 1\})}{a_0} \right\rceil^{1/2}.$$

The probability that any one coordinate exceeds B in absolute value is at most $B_0 e^{2G - a_0 B^2}$, also conditionally on previous layers. A union bound gives probability at most $1/4$ that any incoming coordinate exceeds B. The actual internal weights divide the raw coordinates by $\sqrt{n}$ and so satisfy the same bound. Intersecting this event with the covariance-approximation event leaves probability at least $1/2$. Fix a realization in the intersection. Both n and $\max\{B, 4K/n\}$ are bounded by polynomials in d, K, G at fixed depth. The stated margin follows from $7/8 - 1/8 = 3/4$. □

Lemma C.3 (Counting finite realizations)**.** *Fix L, width n, and an entrywise bound $B \ge 1$. The Boolean functions having depth-L, width-n implementations with margin at least $3/4$ and weights bounded by B number at most*

$$\exp\{P \log(3 + 16B\Lambda)\}, \qquad P = n(d+1) + (L-2)n^2 + n, \quad \Lambda = (d + Ln)[1 + n(B+1)]^{L-1}.$$

Proof. Round every weight to a multiple of $(8\Lambda)^{-1}$. The first hidden layer's coordinatewise error is at most $(d+1)/(8\Lambda)$. Every subsequent hidden layer has error at most $n(B+1)$ times the preceding error plus $n/(8\Lambda)$; the same estimate applies at the linear readout. Induction bounds the final error by $1/8$. Each rounded coordinate has at most $3 + 16B\Lambda$ possible values, so the P-parameter grid has the stated size. Rounding preserves every represented Boolean sign function. □